

ADVANCED

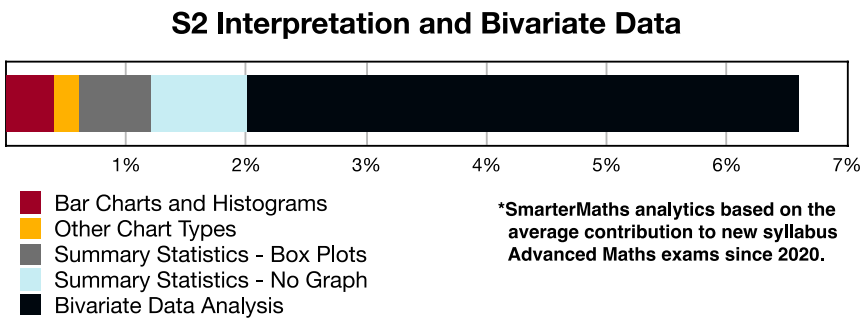
Stage 6

Statistics (Adv), S2 Interpretation and Bivariate Data (Adv)

Summary Statistics - No graph (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 69 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- S2 Interpretation and Bivariate Data* has contributed an average of 6.6% per new syllabus Advanced exam since 2020.
- We have split the area into six sub-topics for analysis purposes which are: 1-*Classifying Data* (0%), 2-*Bar Charts and Histograms* (0.4%), 3-*Other Chart Types* (0.2%), 4-*Summary Statistics - Box Plots* (0.6%), 5-*Summary Statistics - No Graph* (0.8%) and 6-*Bivariate Data Analysis* (4.6%).
- This analysis looks at *Summary Statistics - No Graph* (0.8%).

HSC ANALYSIS - What to expect and common pitfalls

- Summary Statistics - No Graph* (0.8%) has only been examined in one new syllabus Advanced exam to date, in 2022, where it contributed to two separate questions.
- We note past *Std2* questions have required students to calculate a simple dataset's five-number summary, median, IQR, mean and standard deviation.
- An important skill is calculating mean and median changes when a dataset is adjusted.
- Pitfalls:** The 2019 *Std2* exam tested students on "outlier" calculations after the 2017 *Std2 30a* and 2015 *Std2 27d* exams exposed a lack of understanding in this area.
- Major issues have been encountered in identifying the IQR of a cumulative frequency table, and finding the mean of grouped data, where students must use the "class centres".

Questions

1. Statistics, 2ADV S2 2022 HSC 2 MC

Consider the following dataset.

13 16 17 17 21 24

Which row of the table shows how the median and mean are affected when a score of 5 is added to the dataset?

	Median	Mean
A.	Changes	Changes
B.	Stays the same	Stays the same
C.	Changes	Stays the same
D.	Stays the same	Changes

2. Statistics, STD2 S1 2009 HSC 3 MC

The eye colours of a sample of children were recorded.

When analysing this data, which of the following could be found?

- A. Mean
- B. Median
- C. Mode
- D. Range

3. Statistics, STD2 S1 2018 HSC 1 MC

A set of scores has the following five-number summary.

lower extreme = 2
lower quartile = 5
median = 6
upper quartile = 8
upper extreme = 9

What is the range?

- A. 2
- B. 3
- C. 6
- D. 7

4. Statistics, STD2 S1 2004 HSC 6-7 MC

Use the set of scores 1, 3, 3, 3, 4, 5, 7, 7, 12 to answer Questions 6 and 7.

Question 6

What is the range of the set of scores?

- A. 6
- B. 9
- C. 11
- D. 12

Question 7

What are the median and the mode of the set of scores?

- A. Median 3, mode 5
- B. Median 3, mode 3
- C. Median 4, mode 5
- D. Median 4, mode 3

5. Statistics, STD2 S1 2011 HSC 14 MC

A data set of nine scores has a median of 7.

The scores 6, 6, 12 and 17 are added to this data set.

What is the median of the data set now?

- A. 6
- B. 7
- C. 8
- D. 9

6. Statistics, STD2 S1 2011 HSC 17 MC

The heights of the players in a basketball team were recorded as 1.8 m, 1.83 m, 1.84 m, 1.86 m and 1.92 m. When a sixth player joined the team, the average height of the players increased by 1 centimetre.

What was the height of the sixth player?

- A. 1.85 m
- B. 1.86 m
- C. 1.91 m
- D. 1.93 m

7. Statistics, STD2 S1 2013 HSC 14 MC

The July sales prices for properties in a suburb were:

\$552 000, \$595 000, \$607 000, \$607 000, \$682 000, and \$685 000.

On 1 August, another property in the same suburb was sold for over one million dollars.

If the property had been sold in July, what effect would it have had on the mean and median sale prices for July?

- A. Both the mean and median would have changed.
- B. Neither the mean nor the median would have changed.
- C. The mean would have changed and the median would have stayed the same.
- D. The mean would have stayed the same and the median would have changed.

8. Statistics, STD2 S1 2008 HSC 13 MC

The height of each student in a class was measured and it was found that the mean height was 160 cm.

Two students were absent. When their heights were included in the data for the class, the mean height did not change.

Which of the following heights are possible for the two absent students?

- A. 155 cm and 162 cm
- B. 152 cm and 167 cm
- C. 149 cm and 171 cm
- D. 143 cm and 178 cm

9. Statistics, STD2 S1 2008 HSC 8 MC

What is the median of the following set of scores?

<i>Score</i>	<i>Frequency</i>
12	13
14	6
16	2
18	12
Total	33

- A. 12
- B. 13
- C. 14
- D. 15

10. Statistics, STD2 S1 2006 HSC 12 MC

The mean of a set of 5 scores is 62.
What is the new mean of the set of scores after a score of 14 is added?

- A. 38
- B. 54
- C. 62
- D. 76

11. Statistics, STD2 S1 2018 HSC 11 MC

A set of data is summarised in this frequency distribution table.

<i>Score</i>	<i>Frequency</i>
3	1
4	2
5	6
6	7
7	9
8	5
Total = 30	

Which of the following is true about the data?

- A. Mode = 7, median = 5.5
- B. Mode = 7, median = 6
- C. Mode = 9, median = 5.5
- D. Mode = 9, median = 6

12. Statistics, STD2 S1 2007 HSC 21 MC

This set of data is arranged in order from smallest to largest.

5, 6, 11, x , 13, 18, 25

The range is six less than twice the value of x .

Which one of the following is true?

- A. The median is 12 and the interquartile range is 7.
- B. The median is 12 and the interquartile range is 12.
- C. The median is 13 and the interquartile range is 7.
- D. The median is 13 and the interquartile range is 12.

13. Statistics, STD2 S1 2014 HSC 14 MC

Twenty Year 12 students were surveyed. These students were asked how many hours of sport they play per week, to the nearest hour.

The results are shown in the frequency table.

<i>Hours per week</i>	<i>Frequency</i>
0 – 2	5
3 – 5	10
6 – 8	3
9 – 11	2

What is the mean number of hours of sport played by the students per week?

- A. 3.3
- B. 4.3
- C. 5.0
- D. 5.3

14. Statistics, STD2 S1 2016 HSC 21 MC

A grouped data frequency table is shown.

<i>Class Interval</i>	<i>Frequency</i>
1 – 5	3
6 – 10	6
11 – 15	8
16 – 20	9

What is the mean for this set of data?

- A. 6.5
- B. 10.5
- C. 11.9
- D. 12.4

15. Statistics, STD2 S1 2009 HSC 21 MC

The mean of a set of ten scores is 14. Another two scores are included and the new mean is 16.

What is the mean of the two additional scores?

- A. 4
- B. 16
- C. 18
- D. 26

16. Statistics, STD2 S1 2007 HSC 24a

Consider the following set of scores:

3, 5, 5, 6, 8, 8, 9, 10, 10, 50.

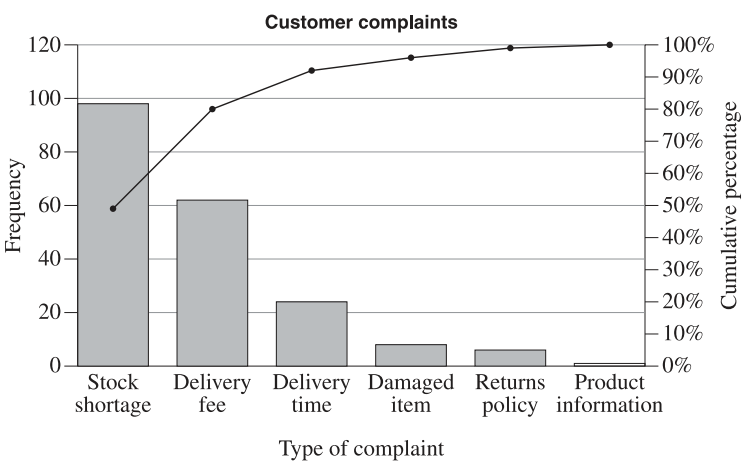
- i. Calculate the mean of the set of scores. (1 mark)
- ii. What is the effect on the mean and on the median of removing the outlier? (2 marks)

17. Statistics, 2ADV S2 2022 HSC 11

The table shows the types of customer complaints received by an online business in a month.

Type of complaint	Frequency	Cumulative frequency	Cumulative percentage
Stock shortage	98	98	49
Delivery fee	62	A	80
Delivery time	24	184	92
Damaged item	8	192	B
Returns policy	6	198	99
Product information	2	200	100
Total	200		

- a. What are the values of A and B? (2 marks)
- b. The data from the table are shown in the following Pareto chart.



The manager will address 80% of the complaints.
Which types of complaints will the manager address? (1 mark)

18. Statistics, STD2 S1 2013 HSC 26b

Write down a set of six data values that has a range of 12, a mode of 12 and a minimum value of 12. (2 marks)

19. Statistics, STD2 S1 2019 HSC 19

The heights, in centimetres, of 10 players on a basketball team are shown.

170, 180, 185, 188, 192, 193, 193, 194, 196, 202

Is the height of the shortest player on the team considered an outlier? Justify your answer with calculations. (3 marks)

20. Statistics, STD2 S1 2017 HSC 27a

Jamal surveyed eight households in his street. He asked them how many kilolitres (kL) of water they used in the last year. Here are the results.

220, 105, 101, 450, 37, 338, 151, 205

- i. Calculate the mean of this set of data. (1 mark)
- ii. What is the standard deviation of this set of data, correct to one decimal place? (1 mark)

21. Statistics, STD2 S1 2015 HSC 27d

In a small business, the seven employees earn the following wages per week:

\$300, \$490, \$520, \$590, \$660, \$680, \$970

- i. Is the wage of \$970 an outlier for this set of data? Justify your answer with calculations. (3 marks)
- ii. Each employee receives a \$20 pay increase.
What effect will this have on the standard deviation? (1 mark)

22. Statistics, STD2 S1 2008 HSC 23f

Christina has completed three Mathematics tests. Her mean mark is 72%.

What mark (out of 100) does she have to get in her next test to increase her mean mark to 73%? (2 marks)

23. Statistics, STD2 S1 2016 HSC 27b

A small population consists of three students of heights 153 cm, 168 cm and 174 cm. Samples of varying sizes can be taken from this population.

What is the mean of the mean heights of all the possible samples? Justify your answer. (2 marks)

24. Statistics, STD2 S1 2017 HSC 30a

A set of data has a lower quartile (Q_L) of 10 and an upper quartile (Q_U) of 16.

What is the maximum possible range for this set of data if there are no outliers? (2 marks)

25. Statistics, STD2 S1 2018 HSC 26e

A cumulative frequency table for a data set is shown.

<i>Score</i>	<i>Cumulative frequency</i>
1	5
2	9
3	16
4	20
5	34
6	42

What is the interquartile range of this data set? (2 marks)

26. Statistics, STD2 S1 2006 HSC 23c*

Vicki wants to investigate the number of hours spent on homework by students at her high school.

She asks each student how many hours (to the nearest hour) they usually spend on homework during one week.

The responses are shown in the frequency table.

<i>Number of hours spent on homework in a week</i>	<i>Frequency</i>
0 to 4	69
5 to 9	72
10 to 14	38
15 to 19	21

What is the mean amount of time spent on homework? (2 marks)

27. Statistics, STD2 S1 2005 HSC 27d

Nine students were selected at random from a school, and their ages were recorded.

Ages		
12	11	16
14	16	15
14	15	14

- i. What is the sample standard deviation, correct to two decimal places? (2 marks)
- ii. Briefly explain what is meant by the term *standard deviation*. (1 mark)

Worked Solutions

1. Statistics, 2ADV S2 2022 HSC 2 MC

Mean decreases.

Median remains 17.

$\Rightarrow D$

◆ Mean mark 48%.

2. Statistics, STD2 S1 2009 HSC 3 MC

Eye colour is categorical data

$\therefore$ Only the mode can be found

$\Rightarrow C$

3. Statistics, STD2 S1 2018 HSC 1 MC

Range = upper extreme – lower extreme

= 9–2

= 7

$\Rightarrow D$

4. Statistics, STD2 S1 2004 HSC 6-7 MC

Question 6

Range = High–Low
= 12–1
= 11
 $\Rightarrow C$

Question 7

9 scores

$\therefore \text{Median} = \frac{9 + 1}{2}$
= 5th score
= 4
Mode = 3
 $\Rightarrow D$

5. Statistics, STD2 S1 2011 HSC 14 MC

Since an even amount of scores are added below and above the existing median, it will not change.
 $\Rightarrow B$

6. Statistics, STD2 S1 2011 HSC 17 MC

Old Mean = $(1.8 + 1.83 + 1.84 + 1.86 + 1.92) \div 5$
= $\frac{9.25}{5}$
= 1.85 m

Since the new mean = 1.86m (given)

New Mean = Height of all 6 players $\div 6$

$\therefore 1.86 = \frac{9.25 + h}{6}$ (h = height of new player)
 $h = (6 \times 1.86) - 9.25$
= 1.91 m
 $\Rightarrow C$

7. Statistics, STD2 S1 2013 HSC 14 MC

Mean increases because new house is sold above the existing average.

Initial median = $\frac{607\,000 + 607\,000}{2} = 607\,000$

New median = 607 000 (4th value in a list of 7)
 $\Rightarrow C$

8. Statistics, STD2 S1 2008 HSC 13 MC

Since the mean doesn’t change

$\Rightarrow 2$ absent students must have a mean height of 160 cm.

Considering each option given,

$(149 + 171) \div 2 = 160$
 $\Rightarrow C$

9. Statistics, STD2 S1 2008 HSC 8 MC

Median = $\frac{n + 1}{2}$
= $\frac{33 + 1}{2}$
= 17th score

$\therefore$ Median is 14
 $\Rightarrow C$

10. Statistics, STD2 S1 2006 HSC 12 MC

Mean of 5 scores = 62

$\therefore$ Total of 5 scores = $62 \times 5 = 310$

Add a score of 14

Total of 6 scores = $310 + 14 = 324$

$\therefore$ New mean = $\frac{324}{6}$
= 54

$\Rightarrow B$

11. Statistics, STD2 S1 2018 HSC 11 MC

Mode = 7 (highest frequency of 9)

Median = average of 15th and 16th data points.

$\therefore$ Median = 6
 $\Rightarrow B$

12. Statistics, STD2 S1 2007 HSC 21 MC

$5, 6, 11, x, 13, 18, 25$

$\text{Range} = 2x - 6$

$25 - 5 = 2x - 6$

$2x = 26$

$x = 13$

$\therefore \text{Median} = 13$

$Q_1 = 6 \quad Q_3 = 18$

$\therefore \text{IQR} = 12$

$\Rightarrow D$

13. Statistics, STD2 S1 2014 HSC 14 MC

Using the class centres

$$\begin{aligned} \text{Total hours} &= (1 \times 5) + (4 \times 10) + (7 \times 3) + (10 \times 2) \\ &= 5 + 40 + 21 + 20 \\ &= 86 \end{aligned}$$

$$\text{Mean hours} = \frac{86}{20} = 4.3$$

 $\Rightarrow B$

◆ *Mean mark 45%*
COMMENT: The mean is calculated using "class centres" in grouped data.

14. Statistics, STD2 S1 2016 HSC 21 MC

Using the centre of each class interval:

$$\begin{aligned} \text{Mean} &= \frac{3 \times 3 + 8 \times 6 + 13 \times 8 + 18 \times 9}{3 + 6 + 8 + 9} \\ &= 12.42 \dots \\ &\Rightarrow D \end{aligned}$$

◆ *Mean mark 43%.*

15. Statistics, STD2 S1 2009 HSC 21 MC

If $\bar{x}$ of 10 scores = 14
 $\Rightarrow \text{Sum of 10 scores} = 10 \times 14 = 140$
With 2 additional scores, $\bar{x} = 16$
 $\Rightarrow \text{Sum of 12 scores} = 12 \times 16 = 192$
 $\therefore \text{Value of 2 extra scores} = 192 - 140$
 $\qquad \qquad \qquad = 52$

$\therefore \text{Mean of 2 extra scores} = \frac{52}{2} = 26$
 $\Rightarrow D$

◆◆◆ *Mean mark 28%.*

16. Statistics, STD2 S1 2007 HSC 24a

i. Total of scores
$$\begin{aligned} &= 3 + 5 + 5 + 6 + 8 + 8 + 9 + 10 + 10 + 50 \\ &= 114 \end{aligned}$$

$$\therefore \text{Mean} = \frac{114}{10} = 11.4$$

ii. Mean
If the outlier (50) is removed, the mean would become lower.

Median
The current median (10 data points)
$$= \frac{5\text{th} + 6\text{th}}{2} = \frac{8 + 8}{2} = 8$$

The new median (9 data points)
$$\begin{aligned} &= 5\text{th value} \\ &= 8 \end{aligned}$$

$\therefore \text{Median will NOT change.}$

17. Statistics, 2ADV S2 2022 HSC 11

a. $A = 98 + 62 = 160$
$$\% \text{ Damaged items} = \frac{8}{200} \times 100 = 4\%$$

Cumulative % after damaged items = 96%
 $B = 92 + 4 = 96$

b. The right hand side cumulative frequency percentage shows that 80% of all complaints received concern stock shortages and delivery fees.
 $\therefore$ The manager will address stock shortages and delivery fees.

18. Statistics, STD2 S1 2013 HSC 26b

12, 12, 12, 16, 18, 24
NB. There are many correct solutions.

19. Statistics, STD2 S1 2019 HSC 19

$Q_1 = 185, Q_3 = 194$

$IQR = 194 - 185 = 9$

Shortest player = 170

Outlier height:

$Q_1 - 1.5 \times IQR = 185 - 1.5 \times 9$
 $= 171.5$

∴ Since $170 < 171.5$, 170 is an outlier.

Mean mark 51%.
COMMENT: The last statement must be made to achieve full marks here!

20. Statistics, STD2 S1 2017 HSC 27a

i. Mean = $(220 + 105 + 101 + 450 + 37 + 338 + 151 + 205) \div 8$
 $= 200.875$

ii. Std Dev = 127.357... (by calc)
 $= 127.4$ (1 d.p.)

♦ Mean mark part (ii) 47%.
IMPORTANT: The *population* standard deviation is required here.

21. Statistics, STD2 S1 2015 HSC 27d

i. 300, 490, 520, 590, 660, 680, 970

Median = 590

$Q_1 = 490$

$Q_3 = 680$

$IQR = 680 - 490 = 190$

Outlier if \$970 is greater than:

$Q_3 + 1.5x \times IQR = 680 + 1.5 \times 190 = \965

∴ The wage \$970 per week is an outlier.

♦ Mean mark (i) 39%.

ii. All values increase by \$20, but so too does the mean.
Therefore the spread about the new mean will not change
and therefore the standard deviation will remain the same.

22. Statistics, STD2 S1 2008 HSC 23f

Total marks in 3 tests

$= 3 \times 72$

$= 216$

We need 4-test mean = 73

i.e. Total Marks (4 tests) $\div 4 = 73$

Total Marks (4 tests) = 292

∴ 4th test score = $292 - 216$

$= 76$

23. Statistics, STD2 S1 2016 HSC 27b

If sample is 1 person,

Possible mean(s): 153, 168 or 174.

♦ Mean mark 40%.

If sample is 2 people,

Possible mean(s): $\frac{153 + 168}{2} = 160.5$

$\frac{153 + 174}{2} = 163.5$

$\frac{168 + 174}{2} = 171$

If sample is 3 people,

Mean: $\frac{153 + 168 + 174}{3} = 165$

∴ Mean of all mean heights

$= \frac{153 + 168 + 174 + 160.5 + 163.5 + 171 + 165}{7}$

$= 165 \text{ cm}$

24. Statistics, STD2 S1 2017 HSC 30a

$IQR = 16 - 10 = 6$

If no outliers,

Upper limit $= Q_U + 1.5 \times IQR$

$= 16 + 1.5 \times 6$

$= 25$

Lower limit $= Q_L - 1.5 \times IQR$

$= 10 - 1.5 \times 6$

$= 1$

$\therefore \text{Maximum range} = 25 - 1$
 $= 24$

♦♦ Mean mark 34%.

25. Statistics, STD2 S1 2018 HSC 26e

42 data points $\Rightarrow$ median $= \frac{21\text{st} + 22\text{nd}}{2}$

$Q_1 = 11\text{th data point} = 3$

$Q_3 = 32\text{nd data point} = 5$

$\therefore IQR = 5 - 3$
 $= 2$

♦♦ Mean mark 27%.

26. Statistics, STD2 S1 2006 HSC 23c*

Hours	Class Centre (x)	Frequency (f)	fx
0 - 4	2	69	138
5 - 9	7	72	504
10 - 14	12	38	456
15 - 19	17	21	357
		Total	1455

MARKER'S COMMENT: This "routine" exercise of finding a mean from grouped data was incorrectly answered by most students! The best responses copied the table and inserted a class-centre column (see solution).

Mean $= \frac{\text{Sum of Scores}}{\text{Total scores}}$
 $= \frac{1455}{200}$
 $= 7.275 \text{ hours}$

27. Statistics, STD2 S1 2005 HSC 27d

i. Sample standard deviation
 $= 1.6914 \dots$ (by calculator)
 $= 1.69$ (to 2 d.p.)

ii. Standard deviation is a measure of how much members of a data group differ from the mean value of the group.